REPORT — Bridge C order3b (shadow localisation by Q-class and characteristic)

PREREG sha256 bf5a02c6…cfdc (unchanged). Run order as preregistered. Scripts are hashed in scripts/SHA256SUMS.

Map 1 (Category A): ray split

Statement. R = R[7/2] + R[595/2], where: - R[7/2] collects the walls ±(0,1), with Q(0,±1) = 7/2; - R[595/2] collects the walls ±(21,−1), with Q = ½(2·441 − 294 + 7) = 595/2.

Proof. The four walls in R are exactly these directions, so this is a partition of a finite sum. ∎

Numerical check: max |R[7/2] + R[595/2] − R| = 2.8e-17 over 216 points, both at x and at −1/x. **Pass.**

Sign control (gating)

- A0: rank 36, closed at w=1.
- Wrong sign: joint rank 72 > 36.
- **Passed.**

Closure tests (Category B; numerical ranks, SVD threshold 1e-9)

group	rank	joint w=½	joint w=1	Θ1 (K2+K3) rank → joint w=½	Θ2 (K4) rank → joint w=½
G[1] = W[1]	90	178	156	62 → 122	29 → 58
G[7/2] = W[7/2]+R[7/2]	60	116	120	40 → 80	20 → 36
G[595/2] = W[595/2]+R[595/2]	100	200	200	67 → 134	33 → 66
G[D] = D	58	116	98	39 → 78	19 → 38
A1	105	210	210	70 → 140	35 → 70

None of these is closed, at w=½ or at w=1, in any block (full values in data/closure3b.json).

Descriptive: A0 restricted to Θ1 gives rank 24, closed; A0 restricted to Θ2 gives rank 12, closed. So the characteristic blocks are S-stable at order 1, and 24 + 12 = 36.

Verdicts

- **P-Q: not localised by Q-class.** All four Q-class groups fail to close.
- **P-Θ: not localised by characteristic.** Neither Θ1 nor Θ2 closes for A1. A1 is full rank in both blocks (70 and 35), and the joint ranks double.

order3b notes (descriptive)

- **What the ranks say.** In every A1 block the joint rank is exactly twice the block rank (70→140, 35→70). In those blocks the S-image of A1 is linearly independent of A1 on this grid. That is the strongest form of non-closure, and it holds per characteristic.
- **A0 splits cleanly.** A0 decomposes as 24 + 12 across Θ1 and Θ2, and each part is S-closed. This is consistent with the vector S-law on the 35-component law vector space, restricted by family.
- **Reading.** At order 2 the obstruction is spread over every Q-class and over both characteristic blocks. Regrouping finite sums by geometric or characteristic labels does not isolate Ξ. A Category B guess (not tested here) is that the correct object is a completed A1: A1 minus an explicit Eichler-type integral term. Its S-defect would be the period of Ξ, and testing that needs a candidate Ξ to be preregistered.
- **Process.** All ranks are numerical. No mod-p certification was done.